ELECTROCHEMISTRY

• Latimer diagram

SUB TOPICS

Never take Approx !

• Nernst Equation

• Emf Of the Cell

• Electrochemical Series

• SRP SOP relation.

• factors affecting Single Electrode Potential

• Cell Notation

• Concentration Cells

• electrolytic cell.

• factors affecting electrolysis. **

• Trick,
  To determine cathode Anode !
  When More than two element Takes Place

• Reversible Cells. which $E^\circ < 1.1\,eV$, $E^\circ > 1.1\,eV$, $E^\circ = 1.1$ cases

• Electrolysis
  (Molten State)

• Electrical resistance / conductance

• Molar Conductivity $\boxed{G^* = \frac{L}{A}}$

• Equivalent Conductivity $\boxed{K = \frac{1}{\rho}}$

• Grothus effect $\boxed{R = \rho \frac{l}{A}}$

• Debye Huckel
  Onsagar.. $G^* G = K$

$\lambda_m = \lambda_m^\circ - (b\sqrt{c})$ $\quad G^* = RK$

• Kohlraush's law. $\boxed{G = \frac{1}{R}}$

$\lambda_m^\circ = \lambda_\infty^\circ + \lambda_a^\circ$ ✓/.

Oxidation : Anode
Reduction : Cathode

$K = G \times \frac{L}{A}$

⇒ Galvanic Cell

Current : Cathode - Anode
flow of $\bar{e}$: Anode - Cathode.

Electrolytic Cell
reverse of it !

⇒ $\text{Solubility} = \sqrt{K_{sp}}$

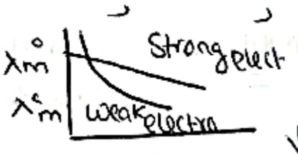
⇒ $\lambda_m = n \times \lambda_{eq}$.

⇒ $\frac{1F}{2} = \frac{wt}{M\,wt}$

⇒ $\text{No. of gram eq} = \frac{Q}{96500}$

⇒ $\eta_{eff} = \frac{i_{act}}{i_{Theoritical}} \times 100$

** Galvanic Cell Anode // Cathode
Electrolytic Cathode // Anode.



Specific resistance
$\rho \rightarrow$ Resistivity : $Ohm\,cm^{-1}$

$G \rightarrow$ Conductance ohm^{-1}, S

$R \rightarrow$ Resistance $Ohm\,cm^{-1}$

$K \rightarrow$ Conductivity (Specific conductance $S\,cm^{-1}$

$\lambda_{eq} \rightarrow$ equivalent conductivity : $Ohm^{-1} cm^{2} eq^{-1}$

$\lambda_M \rightarrow$ Molar conductivity $S\,cm^2\,mol^{-1}$

$G^* \rightarrow$ Cell constant cm^{-1}, m^{-1}

$(\alpha) \rightarrow$ Degree of dissociation

$\alpha = \frac{\Lambda_m}{\Lambda_m^\circ}$ ***

• Faraday's Law's • Batteries

FORMULAS

KEY Concepts

Addition of e^- → Reduction

loSe of e^- → Oxidation

* $E_{cell} = E_{cathode\,(RP)} - E_{Anode\,(RP)}$ & $E_{cell} = E_{OP} + E_{RP}$. $E_{cell\,Max}$ Pore $= E_{cathode\,(RP)} - E_{Anode\,(RP)}$

* SOP $\propto \frac{R.Power}{O.Power} \propto \frac{O.A}{R.A}$ SRP $\propto \frac{O.Power}{R.Power} \propto \frac{R.A}{O.A}$

" Reduction Potential : Strong O.A / Strong RA

$\boxed{\Delta G^o_3 = \Delta G^o_1 + \Delta G^o_2}$

* $\Delta G = -nFE_{cell}$

$\Delta G^o = -nFE^o_{cell}$

$\boxed{\text{SOP} \propto \text{Reactivity}}$

Thermal Stability $\propto \frac{1}{SRP}$

⇨ Enthalpy of cell :

$$\Delta H = -nF\left[E - T\left(\frac{\partial E}{\partial T}\right)_P\right]$$

⇨ Entropy :

$$\Delta S = \frac{\Delta H - \Delta G}{T} = nF\left(\frac{\partial E}{\partial T}\right)_P$$

$\boxed{\eta_{eff} = \frac{\Delta G}{\Delta H} \times 100}$ $E_{cell} = E^o_{cell} - \frac{RT}{nF}\log Q$ * **

Cell Notation : Oxidation // Cathode (Anode) (Reduction) Anode : -ve

/ Phase change Cathode +ve

, Same Phase.

- Basis

Factors affecting Single electrode Potential : * * * *

- conc
- Nature of Metal
- Nature of Electrode
- Pressure Temp

$\frac{W}{Mwt} \times n\,factor = \frac{Q}{96500}$ deposited Mass

" " $= \frac{\eta_{eff} \times i_{The} \times t}{96500 \times 100}$

: E^o_{cell} is Zero when both the electrodes are Made UP of Same Material May have diff concentrations

$$E_{cell_{11}} = \frac{0.0591}{2} \log \frac{P_1}{P_2}$$

"Nernst Equation: when n is constant if not Multiply or divide the two equation !!

$$E_{cell} = E^\circ_{cell} - \frac{0.0591}{n} \log Q$$

- For, Half cell reaction $E_{cell} = E^\circ_{cell} - \frac{0.0591}{2} \log \frac{[Product]}{[reactant]}$
- For, Full cell reaction

$$E_{cell} = -\frac{0.0591}{2} \log \frac{[P]}{[R]}$$

$\Rightarrow E_{RP} = -0.0591[P_H]$

$\Rightarrow E_{OP} = 0.0591[P_H]$ ** Put Stochiometric coeff as Power of Conc

$$E_{cell} = E^\circ_{cell} - \frac{2.303RT}{nF} \log Q.$$

Ex: $A+2B \rightarrow C+D$

$$\boxed{Q = \frac{C}{(B)^2}}$$

At Equilibrium:

$$E^\circ_{cell} = \frac{2.303RT}{nF} \log K_{eq}$$

$$E^\circ_{cell} = \frac{0.0591}{n} \log K_{eq} \quad (\because \text{for } 298K).$$

"Concentration Cells:"
→ Electrode conc cell
→ Electrolyte conc Cell.

** $E_{cell} = -0.0591(P^H_{Cathode} - P^H_{Anode})$

$C_1 > C_2$ Endorganic

$C_2 > C_1$ Exorganic.

E°_{cell} of conc cells $= 0$

RP is intensive Property hence it is not Multiplied or divided ! # Gibb's energy (ΔG) is extensive Prop

E_{cell}(or)

- P_H can be calculated
- Concentrations be calculated
- Pressure can be known

$E_{SHE/NHE} = -0.059 \times P^H$

** $E_{cell} = 0.0591\left[(P_{H_A} - P_{H_C}) + \frac{1}{2} \log \frac{P_A}{P_C}\right]$

//

$$E_{cell} = E^\circ_{cell} - \frac{0.0591}{n}\left[\log\left(\frac{C_1 \text{ Anode}}{C_2 \text{ Cathode}}\right)\right]$$

$E^{\circ}_{cell} = \frac{RT}{nF} \ln K$ ...

* For Solids Reaction Coeff is 1 Also for Pure liquids

* Feasible Means Spontanious ∴ $\boxed{\Delta G = -ve}$

Trick:

Cathod	Anode :
$H_2 > 1\ 2\ 13$ Group	$O_2 > SO_4^{2-}\ NO_3^-\ F^-\ SO_3^{2-}$
$H_2 <$ All Other Elements	$O_2 < Cl^-\ Br^-\ I^-$ $\downarrow Cl_2\ Br_2\ I_2$

* For Calculation of PH Alway take Normality instead of Molarity

: Active Mass (Q) Of Solide is 1.

: if Both electrode are Same then it is Conc cells.

* : Metal Metal insoluable Source Salt Sol = Metal Metal Salt Sol |

$W = \frac{M_{wt} \times Q}{n \times 96500}$ $\boxed{W = ZQt}$

Sopx Reactivity.

$\boxed{\lambda^{\circ}_m = \frac{K_{Salt} \times 1000}{Solubility}}$

PH : $E_{cell} = -0.0591\ PH$.

$P^H_{WA} = \frac{1}{2}[PK_a - \log C]$ $\quad P^H_{WB} = \frac{1}{2}[PK_b - \log C]$

$pH_{\substack{Acidic \\ Buffer}} = PK_a + \log\frac{[Salt]}{Acid}$ $\simeq W_A + W_{ASB} \rightarrow$ Salt

$pOH_{BB} = PK_b + \log\frac{[Salt]}{Base}$ $\simeq W_B + W_{BSA} \rightarrow$ Salt

* $pH_{WASB} = \frac{1}{2}[PK_w + PK_a + \log C]$

* $PH_{WASA} = \frac{1}{2}[PK_w - PK_b - \log C]$

* $PH_{WASWB} = \frac{1}{2}[PK_w + PK_a - PK_b]$

dissociation constant Of weak acid

degree of d

$\alpha = \frac{\lambda^c_m}{\lambda^{\circ}_m} \Rightarrow \boxed{K_a = \frac{C\alpha^2}{1-\alpha}}$ W

* $G = \frac{l}{A}$

$G = \frac{1}{R} = \frac{1}{\rho} \times \frac{l}{A}$

$\boxed{G \times \frac{l}{A} = K}$

$u_e = \frac{Speed}{Electrical\ field} = \frac{Speed}{Potential\ gradient}$

ionic Mobility $(u_e) = \frac{\lambda^{\circ}_m}{96500}$

if $A = 1cm^2$, $l = 1cm$

$\boxed{K = G}$

$\lambda_M = \frac{K \times 1000}{Molarity}$

$\lambda_{eq} = \frac{K \times 1000}{Normality}$

$\boxed{\alpha = \frac{\lambda_{Any\ concentration}}{\lambda_{infinite\ dilution}}}$

$R = \frac{1}{KA}$

$\rho = \frac{1}{K}$ (∵ ρ Specific resistance)

Possiable

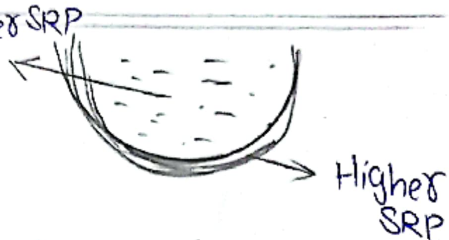
* Reduction Potentials : −3 −2 −1 0 1 2 3 ev

Lower. Higher.

Not Possiable.

Storage in a Bowl or Container : Lower - Higher

Displacement Reaction : Higher with displace lower.



Different type of Electrode:

S.No.	Name of Electrode	Anode	Cathode
1.	Hydrogen Electrode	$Pt(s)\|H_2(g)\|H^+(aq)^2$	$H^+(aq)^2\|H_2(g)\|Pt(s)$
		$E = E^\circ_{cell} - \frac{0.0591}{2} \log \frac{[H^+]^2}{P_{H_2}}$	$\frac{[P_{H_2}]}{(H^+)^2}$
2.	Metal Metal ion electrode	$M(s)\|M^{+n}(aq)$	$M^{+n}(aq)\|M(s)$
3.	Calomel Electrode	$Hg(l)\|Hg_2Cl_2(s)\|Cl^-(aq)^2$	$Cl^-(aq)^2\|Hg_2Cl_2(s)\|Hg(l)$
4.	Redox Electrode	$Pt(s)\|Fe^{+2}(aq)\|Fe^{+3}(aq)$	$Fe^{+3}(aq)\|Fe^{+2}(aq)\|Fe(s)$
5.	Metal in Soluble Salt anion Electrode	$Ag(s)\|AgCl(s)\|Cl^-(aq)$	$Cl^-(aq)\|AgCl(s)\|Ag(s)$

Oxi agent : Sub that oxi other Sub involved reaction by Gaining Accepting e^- //

i. Lowest oxidation state ⇌ Higher O.S

Reducing Agent.